Balancing Generalization and Specialization in Multimodal Retrieval and Multihop QA Systems

Anonymous ACL submission

Abstract

While general-purpose Large Language Models (LLMs) like ChatGPT excel at performing various tasks across different domains, specialized models have proven to outperform their general-purpose counterparts on domain-specific tasks. Recent advances in Retrieval Augmented Generation adapt general-purpose LLMs to specialized domains with prompting and domain-specific fine-tuning. However, for tasks like multimodal question answering (QA), factors that influence one's decision in favoring one approach over the other are underexplored. In this work, we analyze the performance of architectures fine-tuned for multihop multimodal retrieval and question answering on the WebQA benchmark. We compare these to the performance of zero-shot prompting multimodal LLMs, and find that these general purpose LLMs are surprisingly capable of performing both multimodal retrieval and QA. Moreover, we demonstrate that GPT-4o achieves SoTA QA accuracy on WebQA, and is capable of In-Context Retrieval. At the same time, we show that our fine-tuned retrieval and QA models outperform general purpose LLMs in closed domain settings. Finally, we analyze the dependency between retrieval and QA performance. We study the effects of task complexity on model performance and by contrasting zero-shot prompting LLMs with finetuning models. This analysis underscores the overall finding of our work—while zero-shot prompting LLMs exhibit remarkable capabilities in multimodal retrieval and QA, fine-tuned models outperform them for closed domains and high-complexity tasks, emphasizing the need for careful task and domain assessment in model selection.

1 Introduction

Retrieval augmented generation (RAG) with zero-shot prompting and fine-tuning Large Language Models (LLMs) have become the go-to methods for tasks relying on information retrieval and text generation. In many cases the LLMs parametric memory can sufficiently generalize to answer questions without being provided with retrieval mechanisms for out-of-domain knowledge. However, LLMs often hallucinate and provide wrong information in certain scenarios. This problem is amplified even further on open-domain Question Answering (QA) tasks involving multiple modalities. Grounded text generation using retrieved sources (Lewis et al., 2021) has been extensively studied for text-to-text QA tasks, but its application in multimodal settings has not been studied as much.

Our motivation for this research lies in the growing complexity of answering questions using information on the web, where the challenge of navigating the open-domain setting is further complicated by the presence of multiple modalities and sometimes requires reasoning over multiple sources. WebQA is an ideal dataset on which to compare performance of finetuned RAG systems against general purpose LLMs; it is multimodal, with correct answers requiring reasoning over image and text sources. It is multihop, requiring a complex reasoning process over multiple sources. Finally, WebQA questions from different categories can be broken down into subdomains to analyze performance over domains of varying cardinality.

Motivated by the real-world challenges of building retrieval and question answering (QA) systems, we design and fine-tune a closed domain, multimodal, multihop QA model, that is capable of reasoning over a varying number of sources taken as input from an external retriever module. This research contributes to the relatively underexplored domain of multihop reasoning across various input sources and modalities. Our goal is to explore the challenges posed by these scenarios and develop strategies that enable QA models to retrieve relevant information, conduct logical or numerical reasoning across diverse modalities, and generate coherent responses in natural language. To our knowl-

edge, this is the first application of the Fusion-in-Decoder (FiD) architecture (Tanaka et al.; Suhr et al., 2018) that is shown to work with a variable number of inputs, enabling multi-hop reasoning over sources.

In-Context Learning refers to the ability of LLMs to perform any task by simply providing examples in the input prompt (Dong et al., 2022; Min et al., 2022). Inspired by this research, we propose a method to use the LLM itself as a multimodal retriever, potentially eschewing the requirement of a distinct retrieval module, thereby allowing the design of simpler retrieval-augmented QA systems. We dub this method In-Context Retrieval Language Modeling (RLM). To the best of the authors knowledge, In-Content RLM is disparate from other retrieval augmented approaches which utilize external retrieval modules (Ram et al., 2023; Chen et al., b; Liu et al.). Despite being a natural extension of In-Context learning, In-Context RLM has not yet been studied empirically.

Finally, we explore the tradeoff between using zero-shot prompting LLMs and the fine-tuning approach. While we find that, overall, GPT-4o obtains SoTA performance on the WebQA task, outperforming the accuracy of existing fine-tuned RAG approaches by 7%, fine-tuned approaches still perform better on more restricted subdomains[1]. Finally, we validate that GPT-4o is relying on retrieval abilities to solve the task; we find that GPT-4o is capable of retrieving relevant sources in the presence of distractors and furthermore, when GPT-4o fails to retrieve correct sources, it answers incorrectly 75% of the time, meaning that it is not relying on parametric memory for this task.

Contributions Based on our experimentation and analysis on the WebQA benchmark, we make the following contributions:

- Propose a new architecture for multimodal multihop QA that takes variable number of input sources inspired by the Fusion-in-Decoder method.
- Comparison of general purpose LLMs vs specialized models on the WebQA benchmark.
- Observation of In-Context Multimodal Retrieval abilities of GPT-4o and that it does not rely on parametric memory for multimodal QA.
- Analysis of relationship between retrieval and QA task performance.
- Analysis of task and query complexity on the performance of retrieval and QA tasks.

[1]"In-Context RLM" @ https://eval.ai/web/challenges/challenge-page/1255/leaderboard/3168

2 Related Work

2.1 Retrieval Augmented Generation

In this work, we are primarily concerned with retrieval in a multimodal setting, more specifically, a combination of text and image inputs. MuRAG (Chen et al., b) is an end-to-end retrieval and QA model that leverages pretraining on large multimodal corpora to learn a unified embedding space for image and text modalities. It shows promising results on the WebQA and MultiModalQA datasets. However, challenges persist in answering questions related to counting and object recognition, due to visual understanding limitations.

More recently, the Universal Vision-Language Dense Retrieval model (UniVL-DR) showed SoTA performance on the WebQA retrieval task (Liu et al., 2022b). The authors utilize a combination of hard-negative sampling for constrastive learning and augmentation techniques to learn a unified embedding space for multimodal retrieval. In our work, we employ their pretrained model as a retriever and focus on finetuning a QA model that takes the retrieved sources as input before predicting an answer.

In-context Retrieval Augmented Language Modelling (RALM) (Ram et al., 2023) proposes a method to utilize retrieved sources for grounded text generation by simply prepending the sources into the input prompt. This approach provides promising results, and allows the direct use of off-the-shelf LLMs for open-domain QA. We expand upon this idea, adapting it to a multimodal setting, and report our findings.

2.2 Vision Language Models

Our QA model builds on top of a wealth of research on multimodal Vision-Language tasks and multihop language decoders.

FiD Recent work has proposed Fusion-in-Decoder (FiD) architectures for multihop QA (Izacard and Grave; Tanaka et al.). The authors propose concatenating multiple document embeddings together after encoding retrieved passage and question pairs. The concatenation of all passage embeddings is then fed to a decoder.

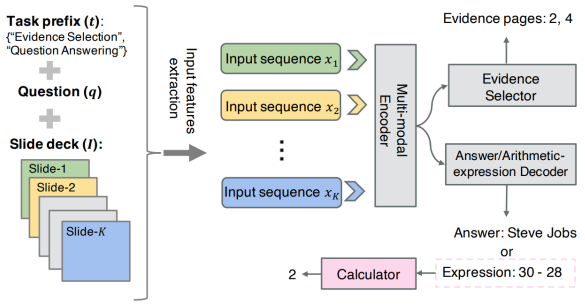


Figure 1: Multihop image-based FiD model (Tanaka et al.)

Similarly, (Tanaka et al.) propose FiD architectures for multihop reasoning over images. They construct a novel dataset out of slide decks and form questions that require combining knowledge from multiple slides to construct the answer. They design a purpose-built FiD model to reason over multiple slides (Figure 1). It is important to note that one limitation of their approach is that they fix the number of inputs to *k sequences* and treat the problem as a *k-hop* multimodal QA task. We propose an FiD model that extends this work to support a variable number of input sequences to overcome this limitation, and tackle multi-hop reasoning tasks.

VoLTA There is also a body of work that develops local alignment methods between captions and images using Graph Optimal Transport (GoT) during training (Chen et al., a; Maretic et al.). GoT is a method for comparing the similarity of two graphs by comparing the Wasserstein distance between signal distributions generated from the graph adjacency matrices. In particular, the VoLTA model generates graph representations of input questions and source images (Pramanick et al., 2022), and incorporates GoT into the loss function. This was shown to reduce embedding distances between entities in the question and source image that have similar relations leading to VoLTA's competitive performance across a range of multimodal tasks.

3 WebQA Task

Modern web searches are multimodal and multihop, integrating various sources and modalities (like text and images) for comprehensive answers. The WebQA dataset is built on this paradigm, requiring multihop reasoning over retrieved images and texts for answer prediction (Chang et al.).

WebQA samples comprise a structured format that includes a question denoted as Q, a collection of m relevant and distractor sources S labeled as s_1 through s_m, and an answer designated as A. Sources come in the form of text snippets or image-description pairs. Image descriptions play a vital role in linking the visual content to the references mentioned in the question because they help identify named entities or geographic details that might not be immediately apparent in the images alone.

The benchmark is designed with a two-step approach in mind; retrieval followed by QA. First, given the question Q and all sources S, we retrieve the set of relevant sources, S'. Using these sources we then generate an answer A'. The following is passed to the QA classifier: $< [CLS], s'_i, [SEP], Q, [SEP] >$ (Chang et al.).

Dataset Upon conducting an initial assessment of the WebQA dataset, we systematically find that all questions (n = 41,732) can be resolved either through text (n = 20,267) or images (n = 21,465), but not a combination of both. Additionally, we find that if a question can be answered using images, it has a maximum of two positive image sources. To investigate the impacts of domain size on model performance, we reformulate the WebQA task as a classification problem and restrict the training and validation data to a subset of the domains present in the original.

Thus, it is possible to simplify the QA problem setting to a prediction task where answers are keywords. The WebQA dataset further breaks down questions requiring one (n = 12,027) or two (n = 9,438) image sources into discrete categories; Yes/No (n = 7,320), color (n = 1,830), number (n = 2,118), shape (n = 565) and choose (n = 4,220). Aside from 'choose', each category has a set number of possible keywords, listed in the appendix. We drop questions in the 'choose'category for this reason.

4 Model Description

4.1 Finetuned QA Model

Multimodal Encoder We select the VoLTA framework as the skeleton for encoding joint text and image representations (Pramanick et al., 2022). VoLTA uses Swin-Base (Liu et al., 2021) and RoBERTa-Base (Liu et al., 2019) as respective visual and textual encoders and we adopt the same encoder choices. Swin Transformer introduced a hierarchical Transformer design with a shifted windows scheme, which computes the self-attention

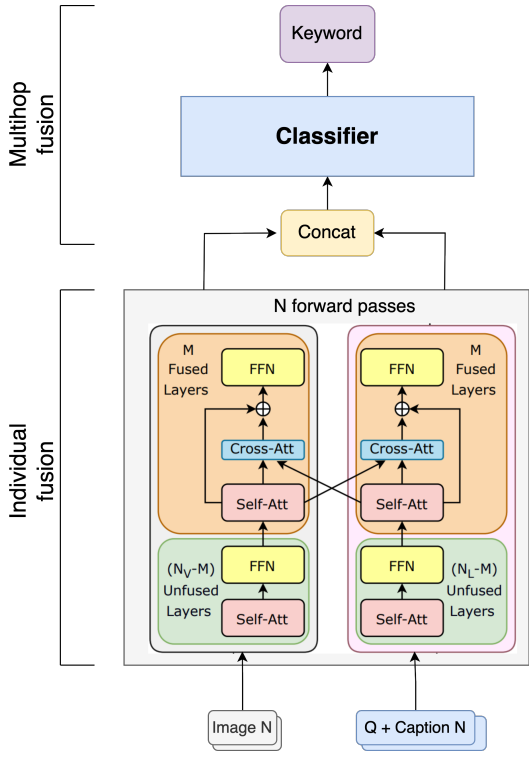


Figure 2: MH-VoLTA: Multihop multimodal VQA model based on VoLTA (Pramanick et al., 2022) encoders

layer over non-overlapping local windows and allows connection by shifting between windows. RoBERTa, known as a robustly optimized BERT, improved over BERT with a few training techniques such as dynamic masking, and explored various NSP training strategies.

Learning VoLTA is pretrained with a combination of objectives. First, Barlow Twin loss L_{BT} is used to reduce redundancy within both image and text pairs and image-text pairs. Then patches of images and tokenized texts are used to construct graphs respectively. Graphs are then aligned using the Graph Optimal Transport (GOT) algorithm. Wasserstein Distance and Gromov-Wasserstein Distance are combined to do node and edge matching. Lastly, Cross-Modal Attention Fusion parameters are pretrained with both masked language modeling L_{MLM} and image-text matching L_{ITM}. The objectives could be categorized to serve three purposes, as shown below:

$$L_{total} = L_{BT} + w_{GOT} * L_{GOT} + L_{MLM} + L_{ITM} \quad (1)$$

Low-Rank Adaptation Adopting the encoder choices as VoLTA, we finetuned the model using Parameter Efficient Fine-tuning techniques. Specifically, we use Low-Rank Adaptation (LoRA) (Hu et al., 2021). Given pretrained weights $W_0 \in \mathcal{R}^{d\times k}$, the parameter update ΔW is represented with a low rank decomposition:

$$W_0 + \Delta W = W_0 + \frac{\alpha}{r} BA \quad (2)$$

where $B \in \mathcal{R}^{d\times r}, A \in \mathcal{R}^{r\times k}, r \ll min(d,k)$. During fine-tuning, W_0 is kept frozen, and only parameters within A and B are trained, reducing the number of parameters required. For our experiments, we set *r=8 and* $\alpha = 32$ for the text encoder and *r=16 and* $\alpha = 16$ for the image encoder, updating only the attention weights.

4.2 Classification on the WebQA Task

For two-image questions, the WebQA fine-tuned VLP baseline (Zhou et al.) takes as input the concatenation of both sources encodings with the query; $< [CLS], s_1, s_2, [SEP], Q, [SEP] >$. As such, it is an extension of VQA model trained on single-hop VQA-2 (Yu et al.), which takes as input $< [CLS], s, [SEP], Q, [SEP] >$. In contrast, FiD implies encoding images with the question so that query-conditioned reasoning is carried out for individual images before a joint reasoning task, resulting in the following input construction:

$$concat(< [CLS], s_1, [SEP], Q >, \\ < [CLS], s_2, [SEP], Q >) \quad (3)$$

We hypothesize that this two-stage reasoning process is essential for the WebQA task, which asks complex questions that require a staged approach. Specifically, we borrowed the 'Fusion-in-Decoder' architecture for questions with two positive image sources. Figure 2 shows the two-stage Multihop VoLTA architecture (MH-VoLTA) where VoLTA's 'fusion-in-backbone' encoders enable multimodal fusion, and our FiD-inspired classifier enables multi-hop fusion between image sources.

We jointly encode each image source returned by the retriever with the query and concatenate the resulting embeddings together before sending them to the MLP classifier to predict the keyword answer label. To handle variable input sequences during classification, we pad single image sources with blank images so that all inputs sent into the classifiers have two images. We fine-tune the models using the AdamW (Loshchilov and Hutter, 2019) optimizer with a learning rate of 0.0001 and a batch size of 32 samples.

4.3 Retrieval and QA on the WebQA Task

Fine-tuned QA We adopt the pretrained UniVL-DR retriever for source retrieval in our finetuned experiments (Liu et al.). Specifically, we embed all text sources, image sources, and queries using UniVL-DR. For each query, we compute cosine similarity between the query and each of the sources, and use the top two ranked image sources and their captions as input to our QA model for fine-tuning as described in Figure 2.

In-Context RALM Previous work in In-Context Retrieval Augmented Language Modeling has demonstrated that LLMs are capable of reasoning over sources without fine-tuning (Ram et al., 2023), and so we compare performance of our fine-tuned classifier against GPT-4o when supplied with relevant sources as retrieved by UniVL-DR. We also compare performance with an oracle retriever, using gold sources provided in the validation set.

In-Context RLM Finally, we hypothesize that In-Context Retrieval Augmented Language Modeling can be extended to what we term In-Context Retrieval (ICR). Here, we test GPT-4o's ability to select relevant sources from a set of distractor sources. We present the extended list of sources to GPT-4o, *without using a retrieval module*, and evaluate its performance on the retrieval task. The prompt used in this case is included in appendix subsection A.1.

5 Methodology

In this section we describe the evaluation metrics, baseline models, and the experimental methodology used for experimentation.

5.1 Experiments

Baseline QA Models We adopt several baseline models from previous works, namely VLP (Zhou et al.), GIT (Wang et al., 2022), GPT-3.5 (Brown et al., 2020), and BLIP-2 (Li et al., 2022). Details of these models are presented in appendix subsection A.2. For the model selection phase, we are only concerned with the one and two image source questions in the WebQA dataset. As such, for all baseline models, we use a source retrieval oracle that provides the positive sources for each question.

Retrieval and QA While we compare performance between the three aforementioned retrieval-QA approaches on the validation image setting due to limitations of our classification model, we evaluate both Zero-shot QA methods on the entire evaluation set.

5.2 Evaluation Metrics

WebQA proposes a composite metric to summarize the performance of generative models; fluency $\times$ accuracy (Chang et al.).

Accuracy Accuracy is the core correctness metric question answering within the WebQA framework. Accuracy is determined by comparing a restricted bag of words (bow) vector between the expected (E) and generated (G) answers, where the generated answer is deemed correct when;

$$\frac{\Sigma \text{bow}_E \text{bow}_G}{\Sigma \text{bow}_E} = 1 \tag{4}$$

To make this measure robust to generative methods answering the same question in different ways, the vocabulary of the vectors is restricted to a specific domain based on the question type; questions are labeled based on these domains which can be yes/no, color, shape, or number. Each category has a pre-defined vocabulary list, given in the appendix.

Fluency WebQA uses "fluency" as one of the question-answering metrics. This fluency metric is assessed using BARTScore (Yuan et al., 2021), a natural language generation (NLG) evaluation measure that focuses on evaluating the quality and coherence of generated text. Note, that for the finetuned model which predicts keyword answers, we ignore fluency and focus on accuracy. As such, accuracy is the main metric of interest, and fluency is given wherever generative methods are used.

Question Complexity Analysis Metrics The analysis of input complexity was conducted using four distinct metrics: Word Count, Flesch-Kincaid Grade Level (Flesch, 2007), and Gunning-Fog Index (Gunning, 1952). All of these metrics were applied to the input question to identify correlations between the complexity of the question to the retrieval recall and QA accuracy (subsection A.5).

6 Results

First, we experiment with multiple QA models to determine which generalized LLMs and specialized, fine-tuned models should be selected for the joint retrieval and QA task. Then, we experiment with different combinations of retrieval systems and chosen QA models for the joint task. Finally,

we analyze how performance on the retrieval task impacts QA accuracy and investigate the relationship between task complexity and performance.

6.1 Model Selection

We explore baseline methods for the QA task on the WebQA validation set. Table 1 gives results for the baseline models. The MH-VoLTA model outperforms all baseline and zero-shot models on the validation set image questions. We also find that MH-VoLTA is capable of reasoning over both one and two-image image questions, and it's performance is on-par with classifiers variants of the classifier trained on one and two image sources separately—These are the '1 img' and '2 img' MH-VoLTA variants in Table 1, see subsection A.3 in the appendix for more details.

GPT-4o and GPT-3.5 models outperform both VLP and GIT baselines. This is particularly impressive for GPT-3.5, which as a unimodal model can only take image caption as input. Despite having mulitmodal inputs, BLIP-2 underperforms GPT-3.5 which we attribute to the difference in decoder quality. We adopt the best performing general purpose LLM (GPT-4o) and specialized VQA model (MH-VoLTA) for all futher experiments.

Table 1: Model selection results on the WebQA validation set. Dataset refers to how the model was finetuned (if applicable). Image set gives QA accuracy on one and two image subsets of the validation set.

Baseline		Metric		# Images	
Model	Dataset	Acc	Flu	1	2
VoLTA	WebQA	**0.71**	–	0.72	0.70
VoLTA	– 1 img	–	–	**0.77**	–
VoLTA	– 2 img	–	–	–	**0.84**
VLP	WebQA	0.50	0.48	0.40	0.42
GIT	VQA-2	0.42	0.19	0.43	0.35
GPT-4o	–	0.564	**0.581**	0.69	0.77
GPT-3.5	–	0.53	0.47	0.41	0.45
BLIP-2	–	0.40	0.20	0.37	0.44

6.2 End-to-End Retrieval + QA

We compare end-to-end performance (retrieval and QA) of the fine-tuned (FT, FT), In-Context RALM (FT, GPT-4o), and In-Context RLM (GPT-4o, GPT-4o) approaches on the WebQA validation set image-based queries. The results show that the fine-tuned UniVL-DR and MH-VoLTA system (FT, FT) outperforms In-Context RLM and RALM (Table 2).

Table 2: QA accuracy and retrieval F1 for each combination of retriever and QA model.

Set	Retriever	QA Model	F1	Acc
Val	None	GPT-4o	–	0.45
Val	GPT-4o	GPT-4o	0.58	0.48
Val	FT	GPT-4o	0.65	0.53
Val	Gold	GPT-4o	1.00	0.56
Val	FT	FT	0.65	**0.69**
Val	Gold	FT	1.00	0.71
Test	GPT-4o	GPT	**0.70**	**0.77**
Test	FT	GPT	0.45	0.73

We also run the generative end-to-end models on the test set, achieving SoTA accuracies.

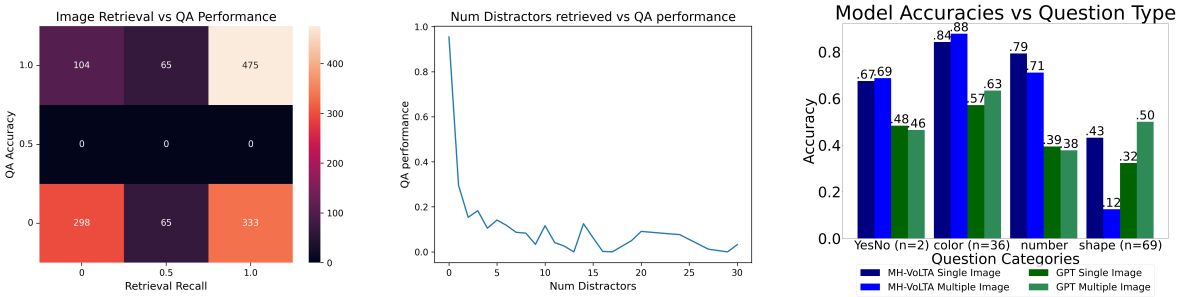
We find that In-Context RALM (FT, GPT-4o) is extremely effective; having 5% higher accuracy than In-Context RLM (GPT-4o, GPT-4o). The UniVL-DR retriever alone achieves an F1 that is 0.07 higher than GPT-4o retrieval, while using a source oracle as a retriever ('Gold') improves QA accuracy with respect to the fine-tuned UniVL-DR retriever by a slim margin; 3.4% for GPT-4o QA and 1.9% for fine-tuned MH-VoLTA QA (Table 2).

We compare In-Context RALM (GPT-4o, GPT-4o) and RLM (FT, GPT-4o) on the WebQA test set by submitting to the leaderboard. Surprisingly, we find that not only does In-Context RLM outperform In-Context RALM accuracy by 4.5%, but that overall it improves upon SoTA accuracy results by 4% at the time of writing.

6.3 Task Analysis

The initial experiments reveal some interesting results; on the validation set UniVL-DR outperforms In-Context Retrieval in terms of both retrieval F1 and QA accuracy, yet on the test set, this is reversed. To understand how retrieval and QA systems interact, we investigate the reliance of the QA task on retrieval correctness. We observe and report interesting relationships between query complexity and retrieval and QA performance.

Impact of Retrieval on QA We find that when GPT-4o correctly retrieves the relevant sources (retrieval recall is high), it has a 59% QA accuracy rate. Conversely, if GPT-4o does not retrieve the correct sources, it is unlikely to answer correctly, with a 74% failure rate (Figure 3(a)). We also find that QA performance drops as the number of retrieved distractors increases, showing that poor retrieval performance adversely affects QA (Figure



(a) GPT-4's performance depends upon its retrieval of correct sources and not its parametric memory.

(b) QA performance drops when distractor sources are retrieved.

(c) Task performance depends on question category, ordered by # labels (n).

Figure 3: Comparison of different QA performance factors.

3(b)). This is to say that the QA task is heavily dependent on retrieval performance.

Question Category We report the mean accuracy per question category for Multihop-VoLTA and GPT-4o in Figure 3(c) using source retrieval oracles. We find that whether GPT-4o or MH-VoLTA fairs better depends on question category, with GPT-4o losing out to MH-VoLTA on all categories except for shape. This could be explained in one of two ways; the more complex a domain (the more labels there are to predict), the worse a specialized, fine-tuned model's performance will be. However, it is more likely that the limited training samples for the shape category in comparison to the other categories is the underlying cause.

Question Complexity We find that the accuracy of In-Context RLM is related to complexity (Figure 4). The more complex the question in terms of word count, Flesch-Kincaid Grade, or Gunning Fog Index, the lower the QA performance (Figure 4). Conversely, increasing query complexity improves GPT-4o's In-Context Retrieval ability, as additional complexity provides information on source relevancy. We note that this opposing relationship between retrieval and QA performance is contrary to the finding that the QA task is heavily dependent on retrieval performance (Figure 3(a)). The impact of query complexity on task performance is strong enough to overcome this general principle.

We extend this analysis to the In-Context RALM and MH-VoLTA systems. Here, given the use of the UniVL-DR retriever, the trend between increasingly complexity and increasing retrieval recall disappears (Figure 5). In addition, QA accuracy for both In-Context RALM and MH-VoLTA does not suffer under increased query complexity. As such, ICR is limited by query complexity, but approaches that utilize fine-tuned retrievers are not.

7 Discussion

We explore the use of generalized language models and specialized fine-tuned models on the WebQA benchmark through model selection on the QA task (Table 1). Our MH-VoLTA model outperformed all baselines, including specialized VQA models fine-tuned on WebQA, and a mixture of unimodal and multimodal language models. We demonstrate that MH-VoLTA is equally capable of performing QA on single and two-image source queries, making it the first multimodal multihop model capable of handling varying numbers of input sources, albeit with limited variance in input length (Figure 3).

Turning to the retrieval task, we compare the UniVL-DR model (Liu et al.) with GPT-4o In-Context Retrieval (Table 2), and find that on the validation set, UniVL-DR outperforms GPT-4o. When our systems are evaluated end-to-end on the combined retrieval and QA tasks, this result holds—the In-Context RALM and MH-VoLTA systems that use the UniVL-DR retriever outperform our In-Context RLM approach with GPT-4o. In addition, we provide a performance upper bound using the 'gold' source oracle and lower bound using no retrieval and relying solely on GPT-4o's parametric memory. As the sophistication of the retrieval system increases, so does the resulting QA accuracy.

This leads us towards a more thorough analysis of the QA task's dependence on the retrieval stage. Unsurprisingly, we find that correct retrieval leads to high QA accuracy (Figure 3(a)), and poor retrieval performance is detrimental to QA accuracy (Figure 3(b)). Crucially, this demonstrates that GPT-4o is not using parametric memory but instead relies on its In-Context Retrieval ability for the WebQA task. To our knowledge, GPT4-o's In-

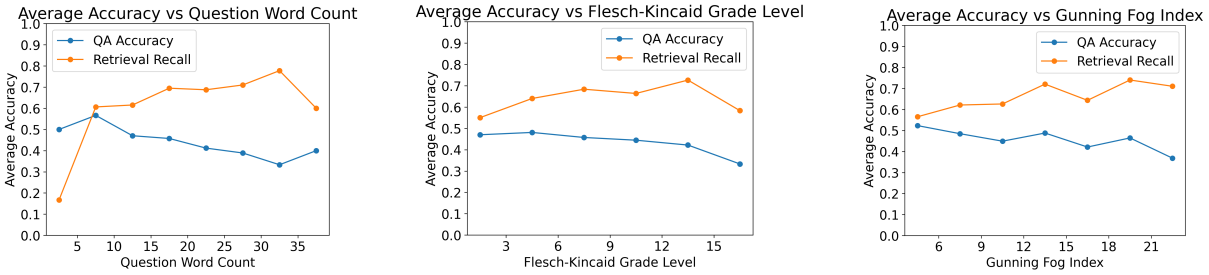


Figure 4: In-Context RLM performance depends upon question complexity; retrieval improves with question complexity, but QA degrades.

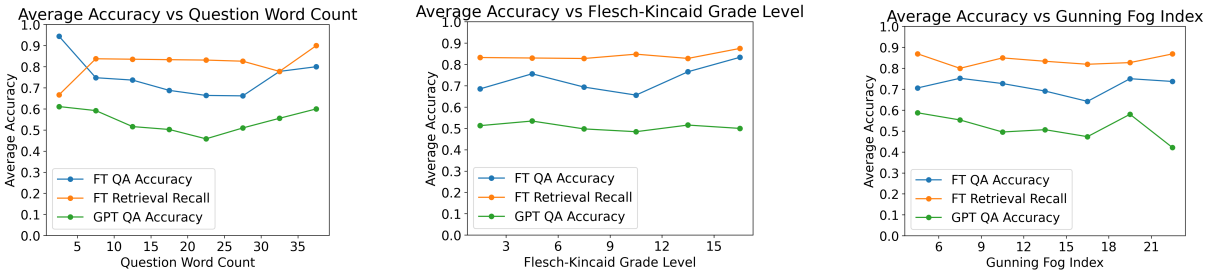


Figure 5: A fine-tuned retriever mitigates the drop in QA performance for In-Context RALM (GPT-4o QA) and fine-tuned QA associated with increased question complexity.

Context Retrieval ability is a novel finding and is an emergent phenomenon in the well-studied space of In-Context Learning (Dong et al., 2022; Min et al., 2022).

We supplement these findings with a more fine-grained exploration of factors that affect retrieval and QA performance. Notably, for MH-VoLTA, it is clear that data availability is important—QA performance is substantially lower for the shape question category, which has by far the fewest training samples (Figure 3). As a result, In-Context RLM is a natural choice in a data-constrained environment. Furthermore, our query complexity analysis reveals that In-Context RLM is most appropriate for low-complexity tasks (Figure 4)—increasing query complexity comes at the cost of In-Context RLM accuracy, so much so that it overcomes the general trend of increased retrieval recall leading to increased QA accuracy. We note that this is not the case for In-Context RALM or MH-VoLTA, where query complexity does not appear to affect retrieval recall or QA accuracy (Figure 5), making these methods preferred where task complexity is unavoidably high.

Finally, we demonstrate that despite the lack of a SoTA fine-tuned retriever, In-Context RLM outperforms In-Context RALM and that In-Context RLM achieves SoTA performance on the WebQA benchmark. This, along with this analysis that points towards where In-Context RLM using GPT-4o, or specialized models like MH-VoLTA may be more applicable, forms the core contribution of this research. We hope that future research on multimodal QA and, more generally, on retrieval in language models, can benefit from this work.

8 Conclusion

In this paper, we compare general-purpose large language models (LLMs) and specialized fine-tuned models on multimodal question-answering (QA) tasks using the WebQA benchmark. Our findings indicate that while zero-shot prompting LLMs like GPT-4o demonstrate impressive capabilities in both retrieval and QA, specialized models such as MH-VoLTA excel in well-defined problem domains. Our analysis underscores the importance of task complexity and data availability in model selection, suggesting that zero-shot prompting LLMs are suitable for simpler tasks and limited data scenarios, and that fine-tuned models are preferable for more complex tasks and data-rich environments. We highlight the role of retrieval accuracy in WebQA performance. Finally, we find that GPT-4o's In-Context Retrieval is notably effective, enabling it to achieve SoTA results on the WebQA benchmark. Our work advances the understanding of retrieval-augmented generation and multimodal QA systems, providing a foundation for future research in this area.

9 Limitations

Our analysis of fine-tuned QA and our Multihop-VoLTA model on the WebQA benchmark is limited by our architecture. In line with previous works on NLVR and VQA tasks, the Multihop-VoLTA model is a classifier. As such, we do not answer questions on the 'text' category, which is the most open domain variant of the benchmark, and instead focus on the multimodal, multihop aspect of questions in categories that specifically require visual reasoning (i.e. counting, color, and shape detection).

In addition, while we find trends that indicate domain cardinality is an important factor in determining whether to adopt fine-tuned or zeroshot models when decided which approach to take on a task, we only explore the WebQA task, and our analysis focuses on the impact of question complexity on retrieval and QA performance. The role of domain cardinality on model selection is an important area for future research, and analysis here should be expanded and repeated on more datasets and tasks.

Finally, we claim that LLMs are capable of In-Context Retrieval based on our finding that GPT-4o can retrieve relevant sources for the WebQA task. However, WebQA pre-selects a list of distractor sources and relevant sources, and so it is clear that GPT-4o's performance on the retrieval part of this task is premised upon the WebQA benchmark construction. The positive sources still need to be present among the list of distractors, and for WebQA, the number of distractor sources is held constant for each question. Further work on more traditional unimodal and multimodal retrieval datasets is needed to generalize the notion of In-Context Retrieval.

Recent work has found that LLMs can predict when they should forgo their parametric memory and use a retriever module to supplement its learned knowledge (Labruna et al., 2024). This work extends the parametric in-domain capabilities of LLMs to handle open-domain tasks by formulating methods where the LLMs predict when to use an IR module, which is the standard in most RAG systems and when to solely rely on parametric memory (Labruna et al., 2024).

10 Ethical Considerations

The implications of this research to the problem of model selection are clear and the potential for misuse of either the models or the complexity analysis is low.

However, we do find that GPT-4o is capable of retrieval. While this seems benign, it lowers the 'barrier of entry' for LLMs to be used to reason over documents that they are not explicitly trained on. This may allow end users of LLMs to circumvent measures taken to remove questionable and / or dangerous content from training material, and so the misuse potential for In-Context Retrieval is large. This warrants future work on building safety benchmarks for LLMs that test the LLMs ability to distinguish safe and unsafe sources provided in prompts for In-Context Retrieval.

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A Appendix

A.1 In-Context Retrieval Prompt

system: Answer the question in one word. Then list the Fact_ID or Image_ID of all facts used to derive the answer in square brackets.
human: Question: <query>
human: Text Facts: [fact_id_1: fact_1, ..., id$_n$: $fact_n$]
human: Image_ID: img_id_1, Caption: img_caption_1
human: [Input_type=image] image_url=url_1
...
human: Image_ID: img_id_m, Caption: img_caption_m
human: [Input_type=image] image_url=url_m

A.2 Baseline Models

VLP The VLP transformer model consists of a unified encoder and decoder (Zhou et al.). The VLP architecture is made up of 12 layers of transformer blocks trained according to the BERT bidirectional and the seq2seq objectives where the self-attention module in the transformer block are defined as;

$$A^l = softmax(\frac{Q^T K}{\sqrt{d}} + M)V^T \tag{5}$$

where $V = W_V^l H^l - 1, Q = W_Q^l H^l - 1, K = W_K^l H^l - 1$. As in (Vaswani et al., 2017), a feedforward layer (with residual) maps A^l to H^l. The model is trained on image caption pairs, and then fine-tuned for the VQA task. Finetuning follows by taking the hidden states from the final layer and feeding them to a multi-layer perceptron. The model used has been fine-tuned twice, once on the VQA dataset (as described by (Yu et al.)), and again on the WebQA dataset.

GIT To contrast with VLP, a pretrained multihop VQA model, we use a pre-trained Generative Image-to-Text Transformer (GIT) (Wang et al., 2022). GIT employs a simplified VQA architecture with one encoder for images and one decoder for text. As such, the model is explicitly incapable for multihop VQA between text and images, so it serves as a baseline for pre-trained models that do not utilize image descriptions, and so we concatenate image sources if there are more than one.

GIT is pre-trained using the language modeling task (as opposed to MLM which is used by VLP) where the model learns to predict captions in an auto-regressive manner. For VQA fine-tuning, the text input is swapped to the query, so that answers are predicted.

BLIP-2 Similar to VLP, the Bootstrapping Language-Image Pre-training model (BLIP) is a unified vision language pre-trained model (Li et al., 2022). It relies on a visual transformer which is less computationally demanding and is pre-trained on over 100 Million image-caption pairs using a contrastive loss (ITC) for image-text contrastive alignment and image-text matching (ITM). In addition to the ITC and ITM losses, the authors introduce an additional Image-grounded text generation (ITG) loss that trains the Q-former encoder to generate texts, given input images as the condition. The final objective to minimize is defined as;

$$Loss = \text{ITC} + \text{ITG} + \text{ITM} \tag{6}$$

GPT-3.5 Turbo Throughout the dataset, a consistent challenge emerges: the model must focus on details, understand them, and accurately respond to questions, even after the provision of positive source images. This challenge has led to the exploration of an image-to-text approach, where the task involves generating descriptive captions for the images. This transforms the problem into a unimodal text retrieval and generation task. Using this method, the SOLAR model has had success on the WebQA task (Liu et al., 2022a). Accordingly, we include a zero-shot oracle baseline, passing queries and image captions to gpt-turbo-3.5 (Brown et al., 2020).

A.3 Multihop VoLTA on one vs two image sources

The results for finetuning VoLTA on the WebQA dataset experiments are provided in Table 3. Here, we operate under the assumption that there exists an oracle capable of supplying us with positive image sources.

We find that the distribution of question types is very different between questions with one and two image sources. Questions with two image sources have more questions in categories with higher random accuracy like Choose and YesNo, whereas questions with one image source have more questions in the harder categories like shape, color, and number, where random accuracy is much lower given the larger answer domain.

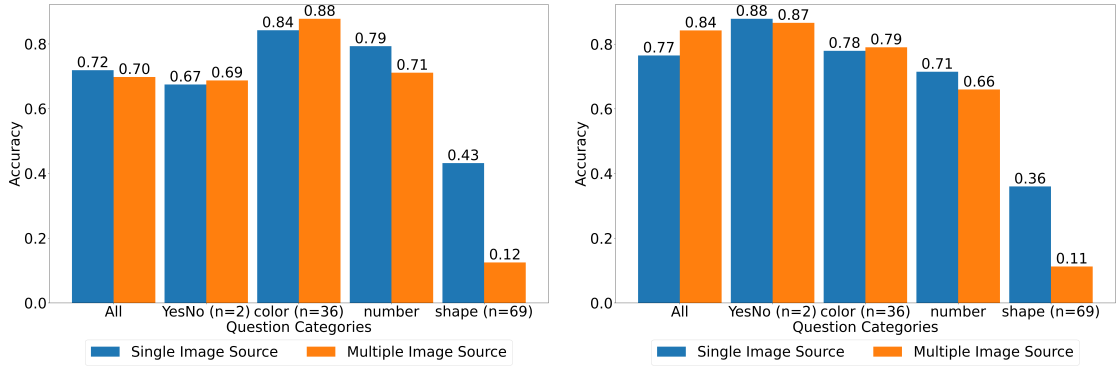
As such, we explored the application of Multihop-VoLTA in addressing queries based on single images, questions involving two images, and a combination of both single and two-image queries (referred to as multiple images). On the evaluation set, we had 760 single image queries and 576 two image queries. We find that the variable Multihop-VoLTA model is en-par with the less flexible models.

Table 3: MH-VoLTA results and dataset breakdown

	No. of Samples	Accuracy
Single Image	760	0.764
Two Images	576	0.851
Multiple Images	1336	0.799

The results presented in 6(b) demonstrate the robustness of our model when trained independently for single-image and two-image questions. 6(a) illustrates a comparable graph for joint fine-tuning of VoLTA on both single-image and two-image sources. Interestingly, there is virtually no distinction in accuracy, even up to the second decimal place, between individual and joint training. This underscores the stability of our fine-tuning approach for VoLTA across both training paradigms. The MH-VoLTAmodels have on the order of 100M parameters, of which 10M are trainable after applying LoRA. All models are trained for 50 epochs on a Nvidia A6000 with 48Gb VRAM.

Moreover, upon examining question categories, our model exhibits consistent performance for both single-image and two-image questions in both training regimes. The similarity in results across different question categories further reinforces the reliability and stability of our model's performance.



(a) Performance of the variable MH-VoLTA classifier, trained on single and two-image sources.

(b) Performance of separate single and two-image MH-VoLTA classifiers variants.

Figure 6: Comparison of the variable MH-VoLTA model (left) vs one and two image models (right) across different question categories, split by the number of image sources per question.

A.4 Question Category Domain Lists

```
yesno_set = {'yes', 'no'}
```

```
color_set = {
    'orangebrown', 'spot', 'yellow', 'blue', 'rainbow', 'ivory',
    'brown', 'gray', 'teal', 'bluewhite', 'orangepurple', 'black',
    'white', 'gold', 'redorange', 'pink', 'blonde', 'tan', 'turquoise',
    'grey', 'beige', 'golden', 'orange', 'bronze', 'maroon', 'purple',
    'bluere', 'red', 'rust', 'violet', 'transparent', 'yes', 'silver',
    'chrome', 'green', 'aqua'
}
shape_set = {
    'globular', 'octogon', 'ring', 'hoop', 'octagon', 'concave', 'flat',
    'wavy', 'shamrock', 'cross', 'cylinder', 'cylindrical', 'pentagon',
    'point', 'pyramidal', 'crescent', 'rectangular', 'hook', 'tube',
    'cone', 'bell', 'spiral', 'ball', 'convex', 'square', 'arch', 'h',
    'cuboid', 'step', 'rectangle', 'dot', 'oval', 'circle', 'star',
    'crosse', 'crest', 'octagonal', 'cube', 'triangle', 'semicircle',
    'domeshape', 'obelisk', 'corkscrew', 'curve', 'circular', 'xs',
    'slope', 'pyramid', 'round', 'bow', 'straight', 'triangular',
    'heart', 'fork', 'teardrop', 'fold', 'curl', 'spherical',
    'diamond', 'keyhole', 'conical', 'dome', 'sphere', 'bellshaped',
    'rounded', 'hexagon', 'flower', 'globe', 'torus'
}
```

A.5 Question Complexity Analysis Metrics

The Flesch-Kincaid Grade Level is a readability metric that evaluates the difficulty of a text based on the length of its words and sentences.

The Gunning Fog Index is a readability test used in linguistics to assess the complexity of English writing.